

Data Visualization and Interpretation Lab – PROJECT REPORT

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PROGRAMME: B.TECH. CS - DATA SCIENCE



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Title: Online Retail Data Analysis and Customer Segmentation

Introduction

This project analyses a UK-based online retail dataset. It contains over 500,000 invoice lines from 2010 to 2011, with fields such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

The goal is to clean and preprocess the data, conduct exploratory data analysis (EDA), create a customer segmentation and clustering model based on RFM, and share insights through interactive dashboards to help with marketing and strategy decisions.

Dataset and Preprocessing

The original dataset includes canceled invoices, negative quantities, zero or negative prices, and missing customer identifiers.

Data cleaning steps removed rows without CustomerID, excluded negative quantities and non-positive UnitPrice values, and eliminated known cancellation records to keep only valid completed transactions.

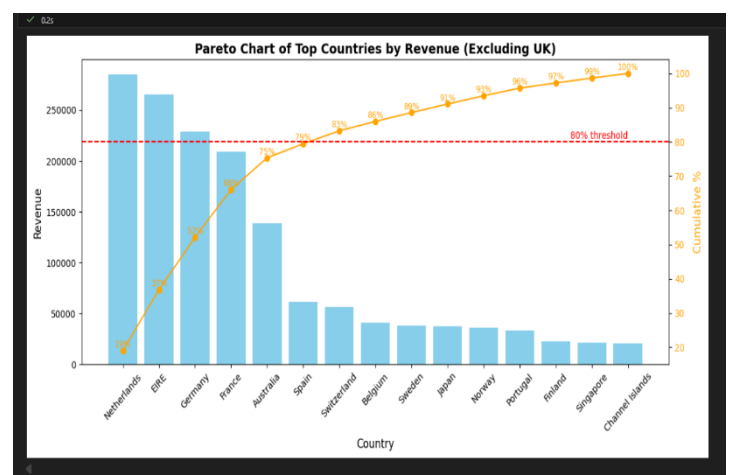
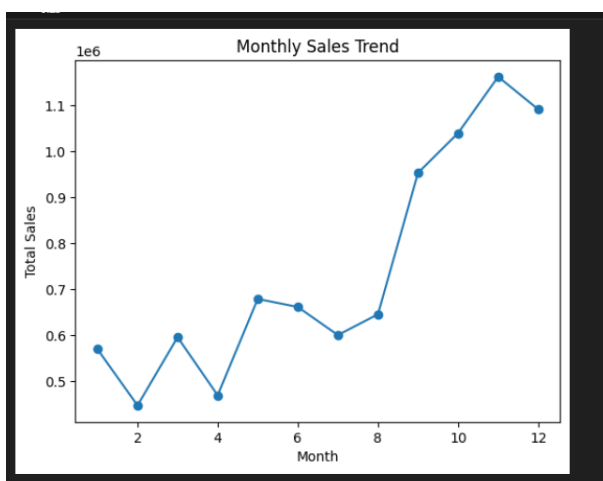
A new variable, TotalPrice, was created by multiplying Quantity by UnitPrice, and temporal features—Year, Month, Day, and Hour—were extracted from InvoiceDate to aid in time-series and seasonal analysis. After cleaning, about 400,000 records remained for analysis.

Exploratory Data Analysis

Univariate and bivariate EDA show that sales are highly concentrated in the United Kingdom, with additional important markets including the Netherlands, Eire, Germany, and France, which together contribute a substantial share of total revenue.

Monthly sales trends show clear seasonality. Sales start lower at the beginning of the year, rise steadily, and reach their highest point in November and December due to strong year-end and holiday demand.

Descriptive statistics indicate that typical orders involve relatively small quantities (median around 6 units) and low unit prices (median below 2), while a small number of large orders and expensive products generate extremely high transaction values. Because the data is skewed, simple averages do not give a full picture. This makes it important to use segmentation to tell apart high-value and low-value customers and products.



RFM Analysis and Rule-Based Segmentation

Customer-level Recency, Frequency, and Monetary (RFM) metrics were computed on the cleaned dataset by aggregating transactions for each CustomerID.

Recency measures the number of days since the last purchase relative to a reference date, Frequency counts distinct invoices, and Monetary represents total revenue per customer across the observation period.

```
# Top 10 Best Customers based on RFM and Monetary value

top_10_customers = (
    rfm[rfm['Segment'] == 'Best Customers']
    .sort_values(by=['Monetary'], ascending=False)
    .head(10)
)

top_10_customers[['Recency', 'Frequency', 'Monetary',
                  'R_Score', 'F_Score', 'M_Score']]
```

CustomerID	Recency	Frequency	Monetary	R_Score	F_Score	M_Score
14646.0	2	73	280206.02	4	4	4
18102.0	1	60	259657.30	4	4	4
17450.0	8	46	194550.79	4	4	4
14911.0	1	201	143825.06	4	4	4
14156.0	10	55	117379.63	4	4	4
17511.0	3	31	91062.38	4	4	4
16684.0	4	28	66653.56	4	4	4
14096.0	4	17	65164.79	4	4	4
13694.0	4	50	65039.62	4	4	4
15311.0	1	91	60767.90	4	4	4

For each RFM dimension, quartile cut-offs were used to assign scores from 1 (least desirable) to 4 (most desirable), resulting in R_Score, F_Score, and M_Score that together capture overall customer value.

Business-friendly segments were then created from combinations of these scores, producing groups such as Best Customers (high frequency and monetary with very recent purchases), Loyal Customers (frequent, consistent buyers), At Risk customers (historically valuable but inactive for a long time), and New Customers (recent, low-frequency and low-spend).

The analysis includes approximately 4,000 active customers, and the RFM segmentation reveals that Best Customers and Loyal Customers collectively contribute the majority of total revenue, despite representing a smaller portion of the customer base.

At the same time, New Customers and At-Risk groups provide opportunities for growth and reactivation through targeted marketing interventions.

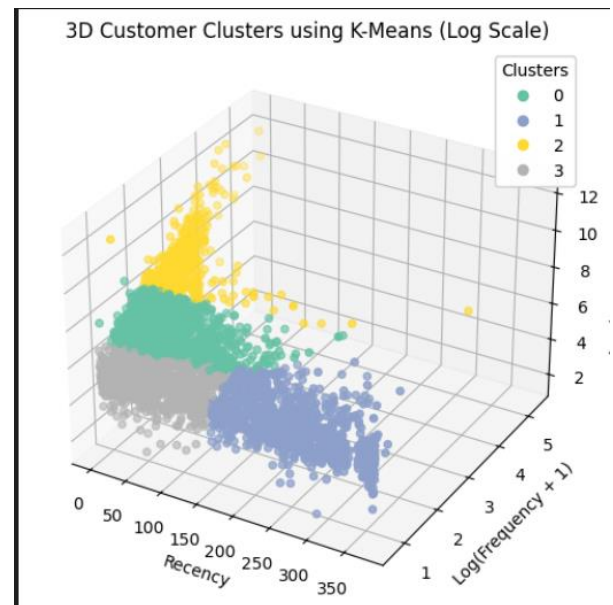
Clustering Model and Evaluation

To uncover more granular patterns beyond rule-based segments, a clustering model was built using the RFM variables.

The Recency, Frequency and Monetary features were first transformed using logarithms where appropriate and then scaled so that all dimensions contributed comparably to distance calculations.

K-means clustering was applied to the transformed RFM data, and the number of clusters was selected by combining inertia (elbow method) and interpretability of resulting groups.

The final solution produced distinct clusters that align with intuitive behaviours such as high-value “champions”, regular mid-value buyers, new or one-time purchasers and high-recency churn-risk customers.

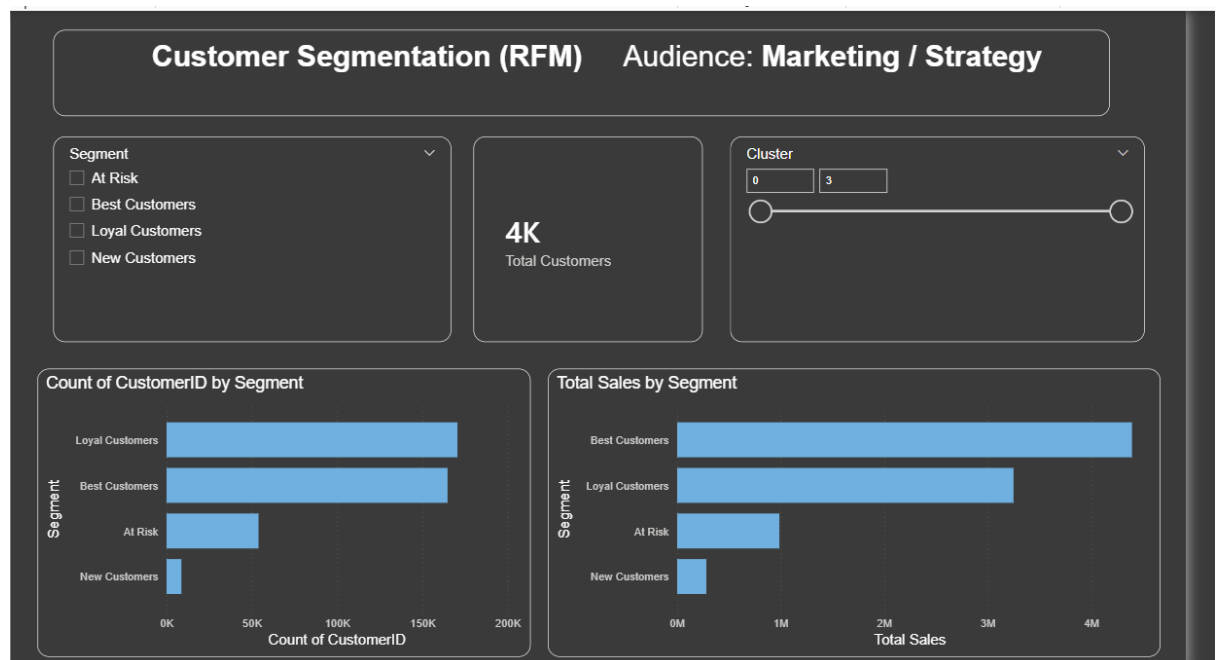
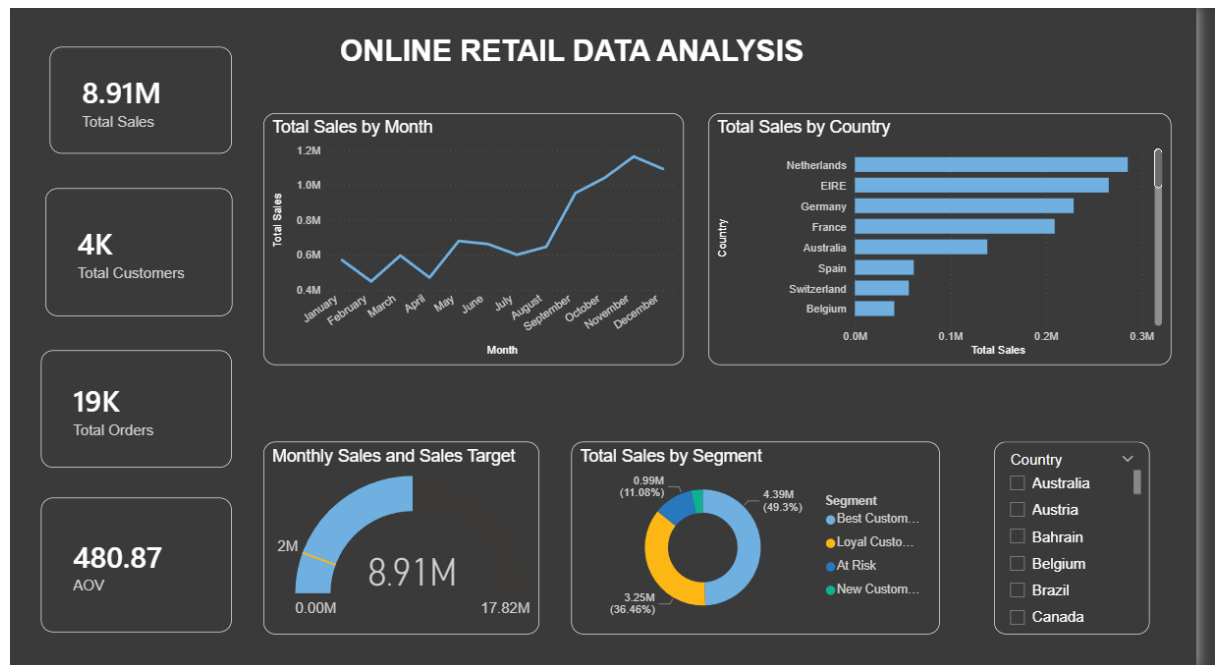


A 3D scatter plot of Recency, $\log(\text{Frequency} + 1)$ and $\log(\text{Monetary} + 1)$, coloured by cluster labels, visually demonstrates the separation between these groups in RFM space and highlights the concentration of revenue in the high-frequency, high-monetary cluster.

Clustering quality was further evaluated using a chi-square test and the silhouette score: the chi-square statistic of 29874.23 with a p-value of 0.0 indicates a significant association between cluster assignments and RFM features, while a silhouette score of about 0.38 reflects a reasonably well-separated clustering structure for business interpretation

Dashboard Design and KPIs (Power BI)

A Power BI dashboard titled “Online Retail Data Analysis” summarises overall business performance using several key performance indicators and trend visuals.



key Insights and Business Recommendations

The analysis confirms that a relatively small share of high-value customers and a handful of key countries drive a disproportionate amount of total sales, indicating that targeted retention and geographic expansion strategies in these areas can yield substantial returns.

Strong seasonality, with pronounced sales peaks in the final quarter, suggests that inventory, staffing and marketing budgets should be ramped up in advance of the holiday period to capture demand efficiently.

Best Customers and Loyal Customers warrant priority treatment through loyalty programmes, exclusive offers and personalised communication, as losing these groups would significantly impact revenue.

At Risk and high-value churn-risk clusters should be targeted with win-back campaigns, including reminder emails, limited-time discounts and product recommendations based on past purchases.

New Customers and lower-value clusters can be nurtured through onboarding journeys, cross-sell bundles and upsell opportunities featuring top-selling products in their region to increase purchase frequency and basket size.

The dashboards enable continuous monitoring of these segments and clusters so that marketing and operations teams can quickly evaluate the impact of campaigns and adjust strategies in near real time.

Conclusion

This project demonstrates an end-to-end analytics workflow for an online retail business, starting from data cleaning and EDA, through RFM scoring and clustering, to dashboard-based visualisation.

The resulting customer segments, model-based clusters and interactive Power BI reports provide a strong foundation for data-driven decisions in marketing, customer relationship management and operational planning, ultimately supporting revenue growth and customer retention.

